



Web Programming
DS362

Project

Deadline: Monday, 4/5/2026 @ 23:59

[Total Mark for this Assignment is 14]

Students Details:

CRN: ###

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ID: S230032898

Instructions:

- You must submit two separate copies (**one Word file and one PDF file**) using the Assignment Template on Blackboard via the allocated folder. These files **must not be in compressed format**.
- It is your responsibility to check and make sure that you have uploaded both the correct files.
- Zero mark will be given if you try to bypass the SafeAssign (e.g. misspell words, remove spaces between words, hide characters, use different character sets or languages other than English or any kind of manipulation).
- Email submission will not be accepted.
- You are advised to make your work clear and well-presented. This includes filling your information on the cover page.
- You must use this template, failing which will result in zero mark.
- You **MUST** show all your work, and text must not be converted into an image, unless specified otherwise by the question.
- Late submission will result in ZERO mark.
- The work should be your own, copying from students or other resources will result in ZERO mark.
- Use **Times New Roman** font for all your answers.

*Learning
Outcome(s):*

2. Create web pages using HTML5 and CSS3.

3. Develop dynamic web pages using JavaScript.

6. Build web applications using PHP and MySQL.

Project

(14 Marks)

In this project, you need to build a website **based on one of the suggested project ideas below**. Your project should be from one of the below ideas. You can work on it individually or as a group of a maximum of 3 members.

Suggested Project ideas

1. **Online Course Registration System:** Home page, Users register/login page, course listing page, course enrollment page, and enrollment history page.
2. **Hotel Booking Website:** Home page, Users register/login page, room search page, room details page, and booking history page.
3. **Online Shopping Website:** Home page, Users register/login page, product categories page, shopping cart page, and order history page.
4. **Clinic Appointment System:** Home page, Users register/login page, doctors listing page, appointment booking page, and appointment history page.

The website must cover all the following aspects of the course (HTML, CSS, JavaScript, PHP, MySQL).

Technology	Description	Mark
HTML	HTML Web elements (header, footer, list, table, form ...)	4
	Navigation (Links are clear and located in the same area on each page. All links are functional)	
CSS	Design (using CSS) Website demonstrates design principles: with consistent alignment, proximity, repetition, and contrast...	3
JavaScript	Form validation, input/output	2
PHP and MySQL	Create and populate a database in MySQL (1-2 tables).	5
	Select records from MySQL database using PHP	
	Insert records into MySQL database using PHP	

	Update records into MySQL database using PHP	
	Delete records from MySQL database using PHP	
Total		14

Important notes:

- The number of Web pages within the website must be **at least 3 pages**.
- The number of dynamic Web pages (using JavaScript) must be **at least 2 pages**.
- The Website must connect to a database (**1 Select + 1 Insert + 1 Update and 1 Delete queries**) and take a screenshot for each one.
- For form validation, at least one popup message must appear after the submit button has been clicked. The message must contain a list of all validation errors on the Web page.
- The project needs to be **submitted in BB** (as an assignment) by **ONE** of the group members. Your instructor will create the submission link.
- Your HTML, CSS, JavaScript, PHP and MySQL codes/scripts that you paste into your answer part of this file must be editable, **not screenshots**.
- You need to take screenshots from each output pages and paste them into your answer in this file.
- Your project must be **original work**. The project will be checked using a **plagiarism tool**. Any matching with an **online project, AI work** or **another/previous student project** will be given **ZERO**.
- **The maximum** number of students in the group is **3**. Only the **group leader** will **upload** the final report on the **blackboard**.

Presentation:

- Each group needs to make a presentation OR a video recording (based on your instructor) of their project. The presentation date will be announced later by your instructor.
- The maximum time for a presentation is **5 mins**.
- In a presentation case, **All** the group members need to **present**.
- In both presentation and video recording cases, **no** need to make **PowerPoint slides**. You need to **run/demonstrate** your project pages and present their functionality and interfaces/pages.

Files to submit/upload:

1. **IT361_Project.docx:**

A word file of the project that contains your answer.

2. **IT361_Project.pdf**

A PDF file of the project that contains your answer.

3. **IT361_Project.zip:**

Contains all the coding files of the project (HTML, CSS, images JavaScript, PHP and MySQL).

*Learning
Outcome(s):*

*2. Create web pages
using HTML5 and
CSS3.*

HTML

{ 4 Marks }

Index.html

```
<?php session_start(); include("includes/setup.php"); ?>
<!DOCTYPE html>
<html><head><title>StayEasy</title><link rel="stylesheet"
href="css/style.css"></head>
<body>

<?php session_start(); ?>

<header class="main-header">
    <div class="logo">StayEasy <span>Hotel</span></div>

    <nav>
        <a href="index.php">Home</a>
        <a href="search_rooms.php">Find Rooms</a>

        <?php if(isset($_SESSION['user'])) { ?>
            <a href="my_bookings.php">My Bookings</a>
            <a href="logout.php">Logout</a>
        <?php } else { ?>
            <a href="login.php">Login</a>
        <?php } ?>
    </nav>
</header>

<section class="hero"></section>
<footer><p>Student Project IT361</p></footer>
</body></html>
```

Login.html

```
<?php
session_start();
include("includes/setup.php");

if(isset($_POST['login'])){
    $u = trim($_POST['username']);
    $p = md5(trim($_POST['password']));

    $res = mysqli_query($conn,"SELECT * FROM users_simple WHERE
username='$u'");

    if(mysqli_num_rows($res)>0){
        $row = mysqli_fetch_assoc($res);

        if($row['password'] == $p){
            $_SESSION['user']=$u;
            header("Location: index.php");
        } else {
            echo "<script>alert('Wrong password');</script>";
        }
    } else {
        echo "<script>alert('User not found');</script>";
    }
}
?>

<html>
<head>
<link rel="stylesheet" href="css/style.css">
<link
href="https://fonts.googleapis.com/css2?family=Poppins:wght@300;400;600&displ
ay=swap" rel="stylesheet">
</head>
<body>

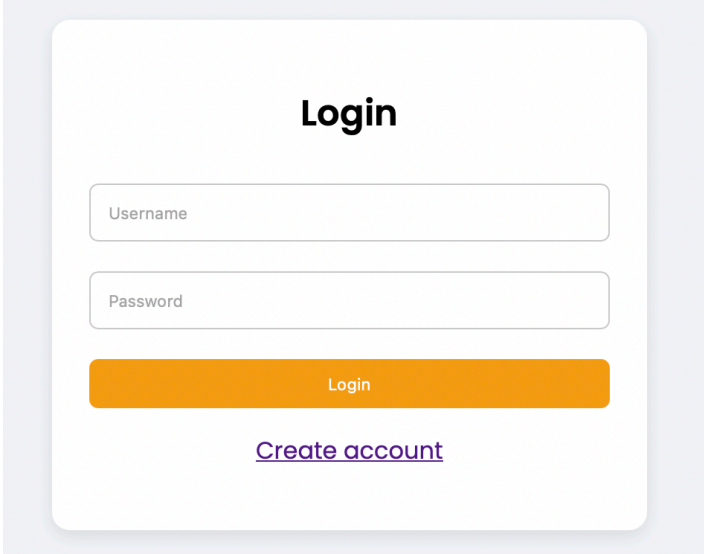
<div class="form-box">
<h2 style="text-align:center;">Login</h2>

<form method="POST">
<input type="text" name="username" placeholder="Username">
```

```
<input type="password" name="password" placeholder="Password">
<button class="btn" name="login">Login</button>
</form>

<p style="text-align:center;">
<a href="register.php">Create account</a>
</p>
</div>

</body>
</html>
```



Login

Username

Password

Login

[Create account](#)

This page allows users to log in using their username and password. It checks the entered data with the database using PHP and displays error messages if the login fails.

Register.html

```
<?php
include("includes/setup.php");

if(isset($_POST['register'])){
    $u = trim($_POST['username']);
    $p = md5(trim($_POST['password']));

    mysqli_query($conn,"INSERT INTO users_simple(username,password)
VALUES('$u','$p')");
    header("Location: login.php");
}
?>

<html>
<head>
<link rel="stylesheet" href="css/style.css">

<link
href="https://fonts.googleapis.com/css2?family=Poppins:wght@300;400;600&displ
ay=swap" rel="stylesheet">

<script>
function validateForm(){
    let u = document.getElementById("user").value;
    let p = document.getElementById("pass").value;

    if(u=="" || p==""){
        alert("Please fill all fields");
        return false;
    }

    if(p.length < 4){
        alert("Password must be at least 4 characters");
        return false;
    }

    return true;
}
```



```
</script>

</head>

<body>

<div class="form-box">

<h2 style="text-align:center;">Create Account</h2>

<form method="POST" onsubmit="return validateForm()">

<input type="text" name="username" id="user" placeholder="Username">

<input type="password" name="password" id="pass" placeholder="Password">

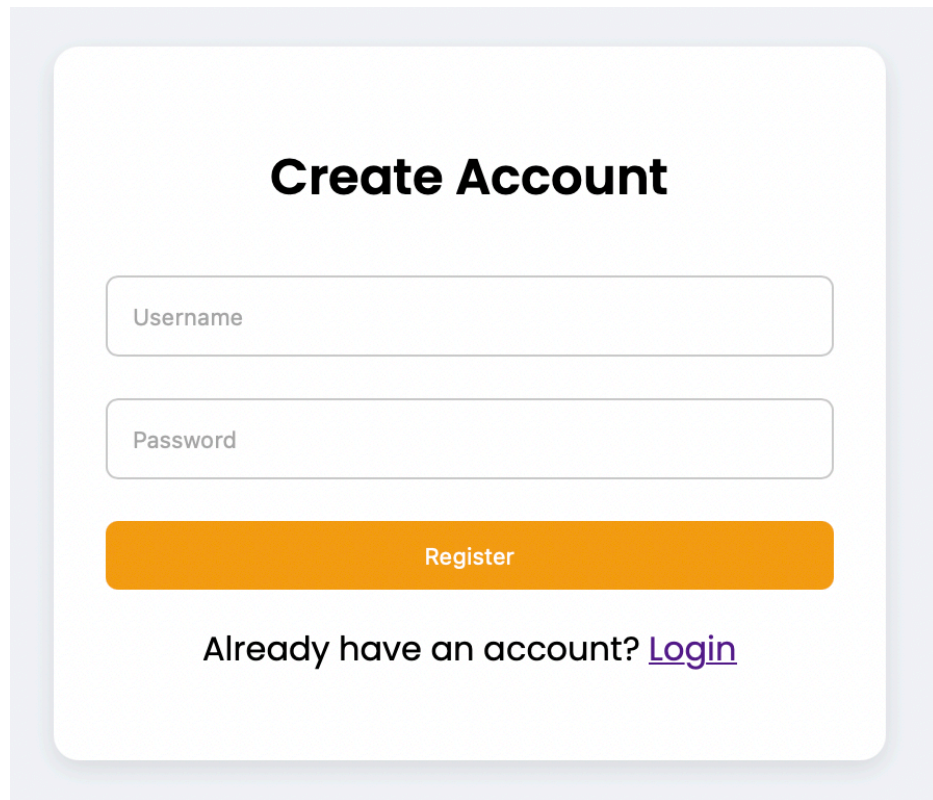
<button class="btn" name="register">Register</button>

</form>

<p style="text-align:center;">
Already have an account? <a href="login.php">Login</a>
</p>

</div>

</body>
</html>
```

A screenshot of a 'Create Account' form. The form is centered on a light gray background. It has a white rounded rectangle containing the title 'Create Account' in bold black text. Below the title are two input fields: 'Username' and 'Password', both with light gray borders and rounded corners. Below these fields is a solid orange button with the text 'Register' in white. At the bottom of the form, there is a line of text: 'Already have an account? [Login](#)' where 'Login' is a purple hyperlink.

Create Account

Already have an account? [Login](#)

This page allows new users to create an account. It uses JavaScript validation to check inputs before sending data to the database.

Seach room page :

```
<?php
include("includes/setup.php");
$res = mysqli_query($conn,"SELECT * FROM hotel_rooms");
?>

<html>
<head>
<link rel="stylesheet" href="css/style.css">
<link
href="https://fonts.googleapis.com/css2?family=Poppins:wght@300;400;600&display=swap" rel="stylesheet">

<script>
function filterRooms(){
    let input = document.getElementById("searchBox").value.toLowerCase();
    let minPrice = document.getElementById("minPrice").value;
    let cards = document.getElementsByClassName("room-card");

    for(let i=0;i<cards.length;i++){
        let text = cards[i].innerText.toLowerCase();
        let price = cards[i].getAttribute("data-price");

        if(text.includes(input) && (minPrice=="" || price >= minPrice)){
            cards[i].style.display = "block";
        } else {
            cards[i].style.display = "none";
        }
    }
}
</script>

</head>

<body>

<?php session_start(); ?>

<header class="main-header">
    <div class="logo">StayEasy <span>Hotel</span></div>
```

```
<nav>
  <a href="index.php">Home</a>
  <a href="search_rooms.php">Find Rooms</a>

  <?php if(isset($_SESSION['user'])) { ?>
    <a href="my_bookings.php">My Bookings</a>
    <a href="logout.php">Logout</a>
  <?php } else { ?>
    <a href="login.php">Login</a>
  <?php } ?>
</nav>
</header>

<div class="page-container">

<h2 class="page-title">Find Your Room</h2>

<div class="filter-box">
<input type="text" id="searchBox" onkeyup="filterRooms()" placeholder="Search
room">
<input type="number" id="minPrice" onkeyup="filterRooms()" placeholder="Min
price">
</div>

<div class="room-grid">

<?php while($row=mysqli_fetch_assoc($res)){ ?>
<div class="room-card" data-price="<?php echo $row['price']; ?>">



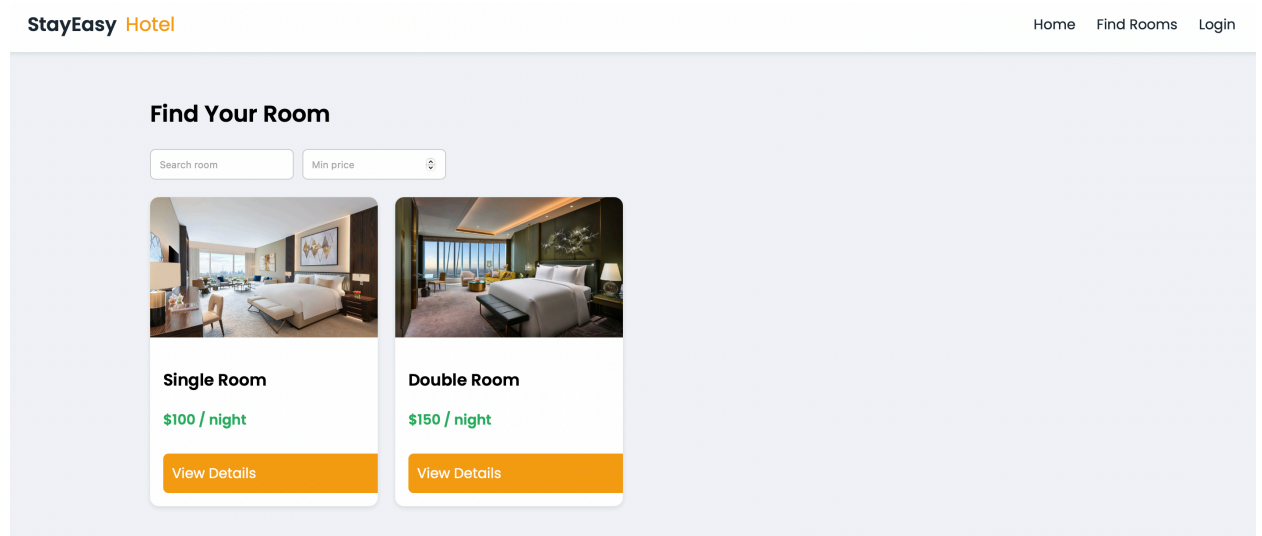
<div class="room-content">
<h3><?php echo $row['room_name']; ?></h3>
<p class="price">$<?php echo $row['price']; ?> / night</p>

<a class="btn" href="room_view.php?id=<?php echo $row['id']; ?>">
View Details
</a>
</div>
</div>
<?php } ?>
```

```
</div>

</div>

</body>
</html>
```



This page displays available hotel rooms from the database. It uses JavaScript (DOM) to filter rooms based on name and price dynamically.

Booking history page :

```
<?php
session_start();
include("includes/setup.php");

$user=$_SESSION['user'];

if(isset($_GET['cancel'])){
    $id=$_GET['cancel'];
    mysqli_query($conn,"DELETE FROM bookings_data WHERE id=$id");
}

$res=mysqli_query($conn,"
SELECT bookings_data.id, hotel_rooms.room_name
FROM bookings_data
JOIN hotel_rooms ON bookings_data.room_id = hotel_rooms.id
WHERE bookings_data.user='$user'
");
?>

<html>
<head>
<link rel="stylesheet" href="css/style.css">
<link
href="https://fonts.googleapis.com/css2?family=Poppins:wght@300;400;600&displ
ay=swap" rel="stylesheet">
</head>
<body>

<?php session_start(); ?>

<header class="main-header">
    <div class="logo">StayEasy <span>Hotel</span></div>

    <nav>
        <a href="index.php">Home</a>
        <a href="search_rooms.php">Find Rooms</a>

        <?php if(isset($_SESSION['user'])) { ?>
```

```
<a href="my_bookings.php">My Bookings</a>
<a href="logout.php">Logout</a>
<?php } else { ?>
<a href="login.php">Login</a>
<?php } ?>
</nav>
</header>

<div class="page-container">

<h2 class="page-title">My Bookings</h2>

<div class="table-box">
<table>
<tr>
<th>ID</th>
<th>Room</th>
<th>Action</th>
</tr>

<?php while($row=mysqli_fetch_assoc($res)){ ?>
<tr>
<td><?php echo $row['id']; ?></td>
<td><?php echo $row['room_name']; ?></td>
<td>
<a class="btn" href="?cancel=<?php echo $row['id']; ?>">Cancel</a>
</td>
</tr>
<?php } ?>

</table>
</div>

</div>
</body>
</html>
```

StayEasy Hotel[Home](#)[Find Rooms](#)[Login](#)

My Bookings

ID	Room	Action
1	Single Room	Cancel
2	Double Room	Cancel

This page shows the user's bookings using a table. It retrieves data from the database and allows the user to cancel bookings (DELETE).

*Learning**Outcome(s):*

2. Create web pages using HTML5 and CSS3.

CSS**{ 3 Marks }**

```
body{
    font-family: 'Poppins', sans-serif;
    margin:0;
    background:#f2f4f8;
}

/* HEADER */
.main-header{
    background:white;
    color:#1c2833;
    display:flex;
    justify-content:space-between;
    padding:15px 30px;
    align-items:center;
    box-shadow:0 2px 5px rgba(0,0,0,0.1);
    position:sticky;
    top:0;
    z-index:1000;
}

/* LOGO */
.logo{
    font-size:22px;
    font-weight:600;
}

.logo span{
    color:#f39c12;
    font-weight:400;
    margin-left:5px;
}

/* NAV LINKS */
nav a{
    color:#1c2833;
    margin-left:20px;
    text-decoration:none;
    font-weight:500;
}
```

```
nav a:hover{
    color:#f39c12;
}

/* CONTAINER */
.page-container{
    padding:30px;
    max-width:1100px;
    margin:auto;
}

.page-title{
    font-size:26px;
    margin-bottom:20px;
}

/* FILTER */
.filter-box{
    display:flex;
    gap:10px;
    margin-bottom:20px;
}

.filter-box input{
    padding:10px;
    border:1px solid #ccc;
    border-radius:6px;
}

/* GRID */
.room-grid{
    display:grid;
    grid-template-columns:repeat(auto-fill, minmax(250px,1fr));
    gap:20px;
}

/* CARD */
.room-card{
    background:white;
    border-radius:10px;
    overflow:hidden;
```

```
    box-shadow:0 2px 6px rgba(0,0,0,0.1);
    transition:0.3s;
}

.room-card:hover{
    transform:translateY(-5px);
}

.room-card img{
    width:100%;
    height:160px;
    object-fit:cover;
}

.room-content{
    padding:15px;
}

.price{
    color:#27ae60;
    font-weight:bold;
}

/* BUTTON */
.btn{
    display:inline-block;
    margin-top:10px;
    padding:8px 15px;
    background:#f39c12;
    color:white;
    border-radius:5px;
    text-decoration:none;
    border:none;
    cursor:pointer;
}

.btn:hover{
    background:#d68910;
}

/* BACK BUTTON */
.back-btn{
```

```
display:inline-block;
margin-bottom:20px;
text-decoration:none;
color:#333;
}

.back-btn:hover{
  color:#f39c12;
}

/* FORM */
.form-box{
  width:350px;
  margin:80px auto;
  background:white;
  padding:25px;
  border-radius:12px;
  box-shadow:0 4px 12px rgba(0,0,0,0.1);
}

.form-box input{
  width:100%;
  padding:12px;
  margin:10px 0;
  border:1px solid #ccc;
  border-radius:6px;
}

.btn{
  width:100%;
  padding:10px;
  background:#f39c12;
  color:white;
  border:none;
  border-radius:6px;
  cursor:pointer;
}

.btn:hover{
  background:#d68910;
}
```

```
/* TABLE */
.table-box{
    background:white;
    padding:15px;
    border-radius:10px;
}

table{
    width:100%;
    border-collapse:collapse;
}

td, th{
    padding:10px;
    border-bottom:1px solid #ddd;
}
```

This file controls the design of the website including layout, colors, spacing, buttons, and responsiveness. It ensures consistent styling across all pages.

*Learning**Outcome(s):*

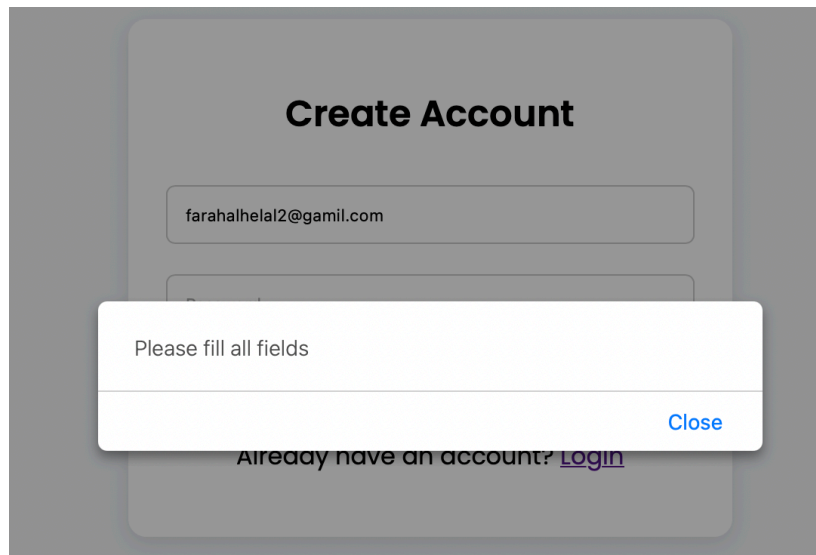
3. Develop dynamic web pages using JavaScript.

JavaScript**{ 2 Marks }**

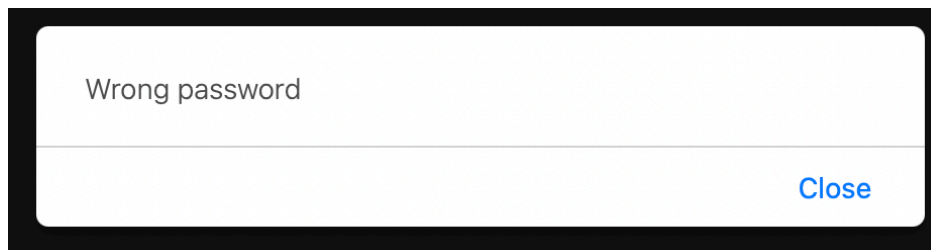
```
// =====  
// REGISTER VALIDATION  
// =====  
function validateForm() {  
    let u = document.getElementById("user").value;  
    let p = document.getElementById("pass").value;  
  
    if(u=="") {  
        alert("Username is required");  
        return false;  
    }  
  
    if(p=="") {  
        alert("Password is required");  
        return false;  
    }  
  
    if(p.length < 4) {  
        alert("Password must be at least 4 characters");  
        return false;  
    }  
  
    return true;  
}  
  
// =====  
// ROOM FILTER (SEARCH PAGE)  
// =====  
function filterRooms() {  
    let input =  
document.getElementById("searchBox").value.toLowerCase();  
    let minPrice =  
document.getElementById("minPrice").value;  
    let cards = document.getElementsByClassName("room-  
card");  
  
    for(let i=0;i<cards.length;i++) {  
        let text = cards[i].innerText.toLowerCase();
```

```
        let price = cards[i].getAttribute("data-price");

        if(text.includes(input) && (minPrice==" " || price
>= minPrice)){
            cards[i].style.display = "block";
        } else {
            cards[i].style.display = "none";
        }
    }
}
```



Register page ERROR



Login page ERROR

This file contains JavaScript functions used for form validation. It checks user inputs and displays alert messages if there are errors.

The project uses DOM (Document Object Model) in JavaScript to interact with HTML elements.

Examples of DOM usage in the project:

1. Accessing form inputs:

```
document.getElementById("regUser").value  
document.getElementById("regPass").value
```

2. Accessing form fields:

```
form.email.value  
form.password.value
```

3. Reading search input:

```
document.getElementById("searchBox").value
```

4. Getting elements:

```
document.getElementsByClassName("room-card")
```

5. Reading content:

```
cards[i].innerText
```

6. Getting attributes:

```
cards[i].getAttribute("data-price")
```

7. Changing display:

```
cards[i].style.display = "none"  
cards[i].style.display = "block"
```

These DOM methods are used to validate forms, filter rooms, and control the display of elements dynamically.

*Learning
Outcome(s):*

*6. Build web
applications using
PHP and MySQL.*

PHP + MySQL

{ 5 Marks }

1. DataBase creation

```
CREATE DATABASE stayerdb;
```

```
USE stayerdb;
```

```
CREATE TABLE users_simple(  
id INT AUTO_INCREMENT PRIMARY KEY,  
username VARCHAR(50),  
password VARCHAR(100)  
);
```

```
CREATE TABLE hotel_rooms(  
id INT AUTO_INCREMENT PRIMARY KEY,  
room_name VARCHAR(100),  
price INT,  
img VARCHAR(100)  
);
```

```
CREATE TABLE bookings_data(  
id INT AUTO_INCREMENT PRIMARY KEY,  
user VARCHAR(50),  
room_id INT  
);
```

Hotel Rooms table

← T →					id	room_name	price	img
Edit Copy Delete	1	Single Room	100	images/room1.jpg				
Edit Copy Delete	2	Double Room	150	images/room2.jpg				

User table

					id	username	password
☐		Edit		Copy		Delete	3 farahalhelal2@gmail.com 4ecc9e9435d30f455c3184de37ffff28

Bookings table

					id	user	room_id
☐		Edit		Copy		Delete	1 1
☐		Edit		Copy		Delete	2 2

The database is created using MySQL and contains three tables: users, rooms, and bookings.

2. SQL Select :

```
$res = mysqli_query($conn, "SELECT * FROM  
hotel_rooms");
```

```
$res=mysqli_query($conn, "  
SELECT bookings_data.id,  
hotel_rooms.room_name  
FROM bookings_data  
JOIN hotel_rooms ON bookings_data.room_id =  
hotel_rooms.id  
WHERE bookings_data.user='$user'  
");
```

3.INSERT Query :

```
mysqli_query($conn, "INSERT INTO  
bookings_data(user,room_id)  
VALUES ('$user', $id)");
```

4. DELET Query :

```
mysqli_query($conn, "DELETE FROM  
bookings_data WHERE id=$id");
```

5. UPDATE Query :

```
mysqli_query($conn, "UPDATE hotel_rooms SET  
price=200 WHERE id=1");
```

These SQL operations are used to manage data:

- SELECT: retrieve rooms and bookings
- INSERT: add new booking
- UPDATE: modify data
- DELETE: remove booking

6.DataBase Connection :

```
$conn =  
mysqli_connect("localhost","root","","staye  
asy_db");
```

This connects the website to the MySQL database using PHP
.mysqli